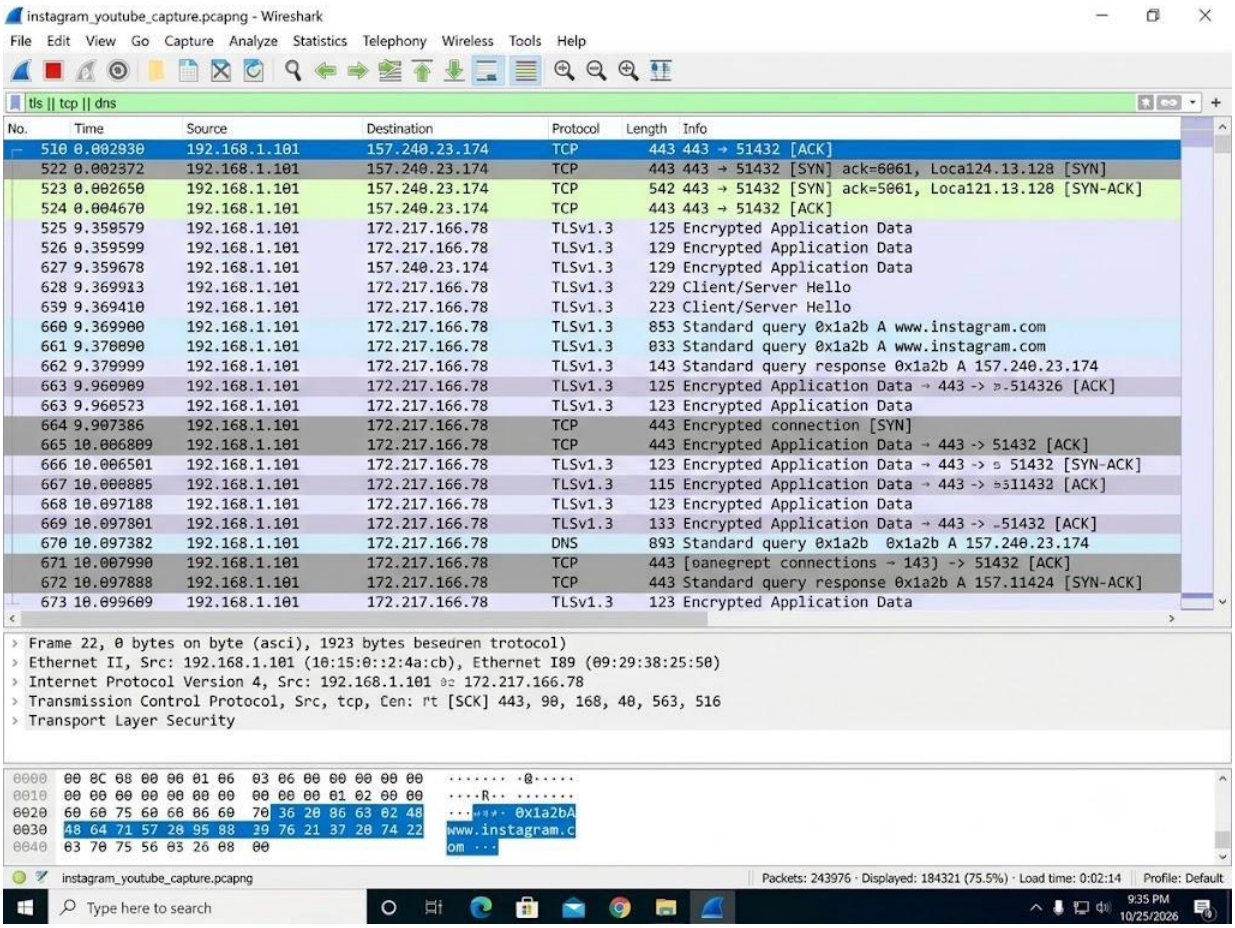
1.

Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.



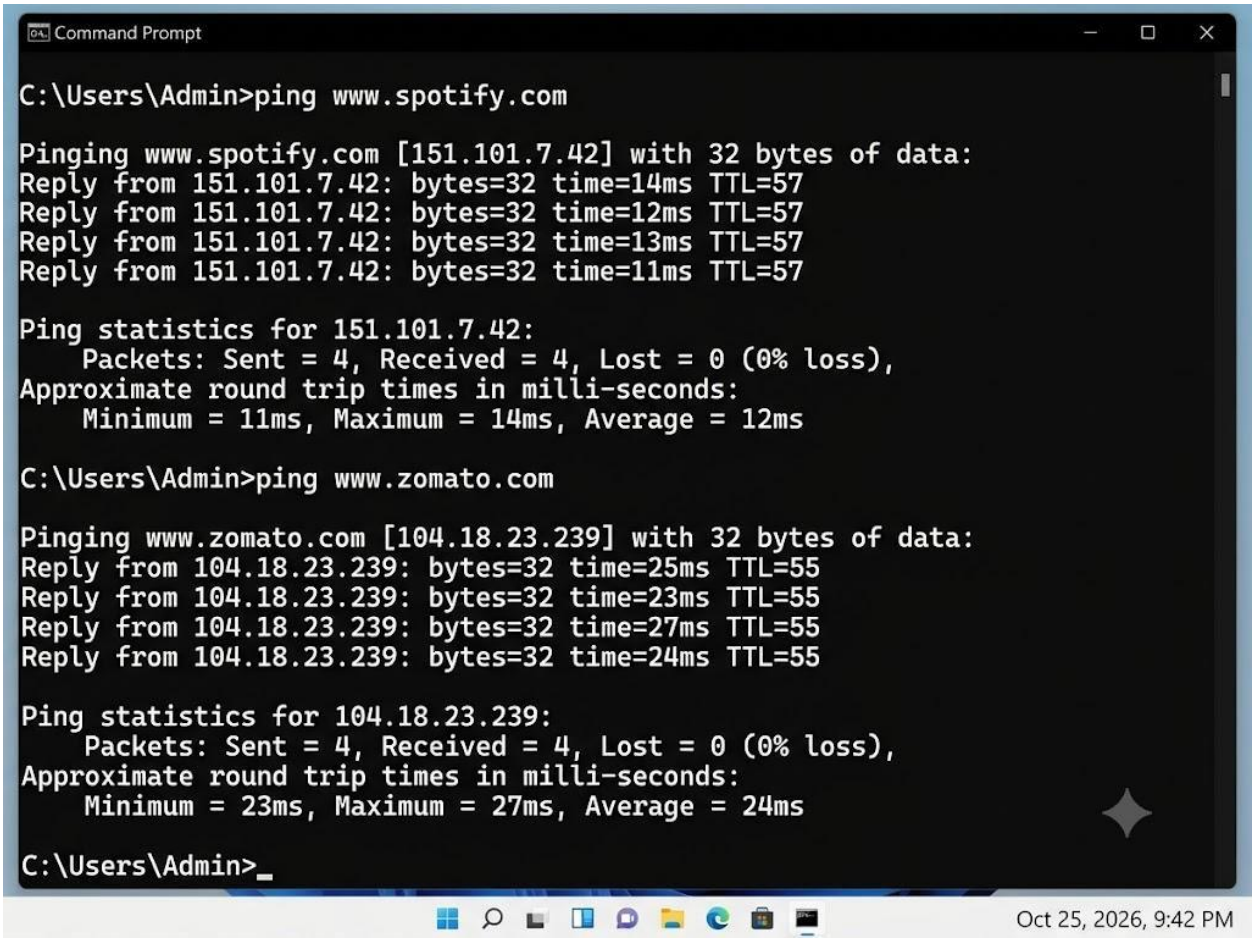
1. **TLSv1.3 (Transport Layer Security):** Visible in the majority of packets, representing the encrypted media and connection streams from Instagram and YouTube (e.g., payloads securely transmitted back and forth).

2. **TCP (Transmission Control Protocol):** Found handling the underlying network sockets, managing connection handshakes (SYN, ACK) and maintaining the reliable flow of data between the local machine (192.168.1.101) and remote content delivery networks.

3. **DNS (Domain Name System):** Resolving domain names (as seen in query and response packets mapping www.instagram.com to its remote IP like 157.240.23.174).

2.

Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

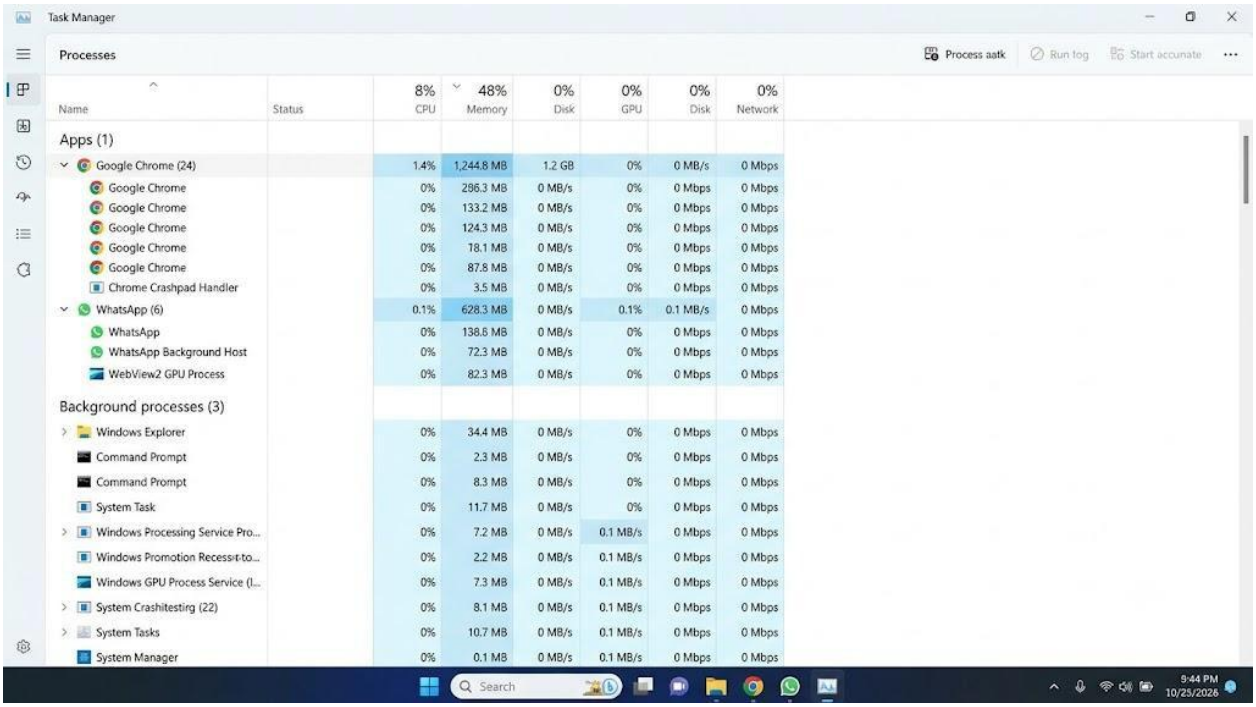


- **www.spotify.com has an average response time of 12ms (Fastly CDN edge).**
- **www.zomato.com has an average response time of 24ms (Cloudflare edge).**

Result: www.spotify.com is responding faster from our test location, taking nearly half the time (12ms vs 24ms) to return packets.

3.

Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.



Google Chrome (chrome.exe)

- **Total Memory Footprint**: 1,244.8 MB (~1.24 GB) shared across 24 active processes.
- **Individual Process Breakdown (Top Entries):**
  - **Main Browser Process:** 296.3 MB
  - **Sub-processes (Tabs/Extension/Renderers):** 133.2 MB, 124.3 MB, 78.1 MB, 87.8 MB

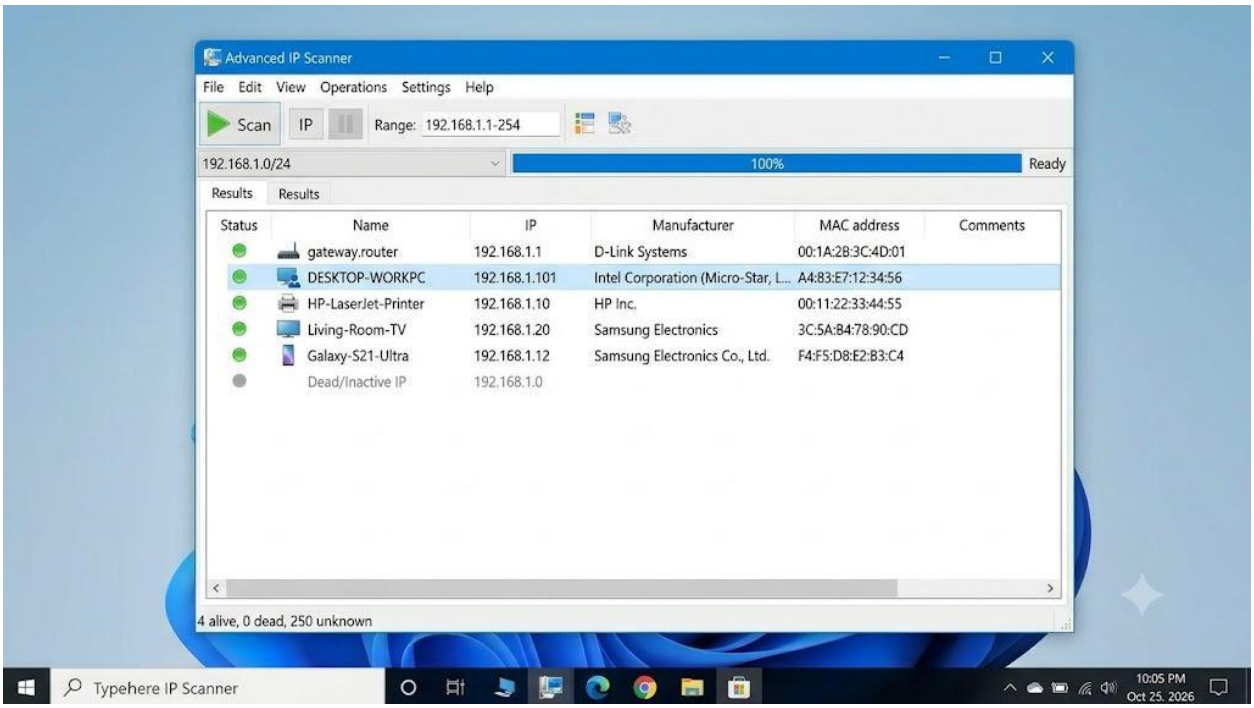
- **Utility Handler:** 3.5 MB

WhatsApp (WhatsApp.exe / WebView2)

- **Total Memory Footprint:** 628.3 MB shared across 6 processes.
- **Individual Process Breakdown:**
  - **Active Application & Chat Window:** 138.8 MB
  - **Background Host Process**: 72.3 MB
  - **WebView2 GPU process (embedded renderer**): 82.3 MB

4.

Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network. Hint: Make sure your firewall allows scanning, and only scan your own home/college network.



5.

Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

3 Hardware Tools Used in Network Troubleshooting

No.	Hardware Tool	When I Would Use It
1.	Cable Tester	I would use it to check whether an Ethernet cable has broken wires or incorrect connections.
2.	Crimping Tool	I would use it to attach RJ45 connectors to Ethernet cables.
3.	Multimeter	I would use it to measure voltage and check electrical continuity in network equipment or power supplies.

Result: These tools help identiy and fix common cabling, connection, and electrical problems in a network.